* Measure of Central Tendency

measure of central Tendency (mean, median mode) are statistical summary tools used to identify the single most representative centre value of data distribution

mean

Population Data (N)

$$\mu = \sum_{i=1}^{N} \frac{x_i}{N}$$

Sample Data (n)

$$\bar{x} = \sum_{i=1}^{n} \frac{x_i}{n}$$

Median

1) Sort all numbers
2) Find central element

eg. $\{1, 2, 3, 4, 5, 2, 1\}$

median $= 4$

$\{1, 2, 3, 3, 4, 2\}$

median $= \frac{3+3}{2} = 3$

Mode

Most Frequent elements

eg $\{1, 2, 5, 5, 5, 4, 5\}$

mode $= 5$

* measure of dispersion

1) Variance

measure of data spread, quantifying how far data points are from the mean by calculating average of squared differences

Low variance = clustered/data points are close to the mean

High variance = Spread out

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^{N} (x_i - \mu)^2$$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

$n-1 \Rightarrow$ Bessel's correction / Degree of Fraction

It corrects the bias that occurs when estimating population variance from sample

	Population	sample
Purpose	To describe population	To estimate population

Standard Deviation

Square root of variance

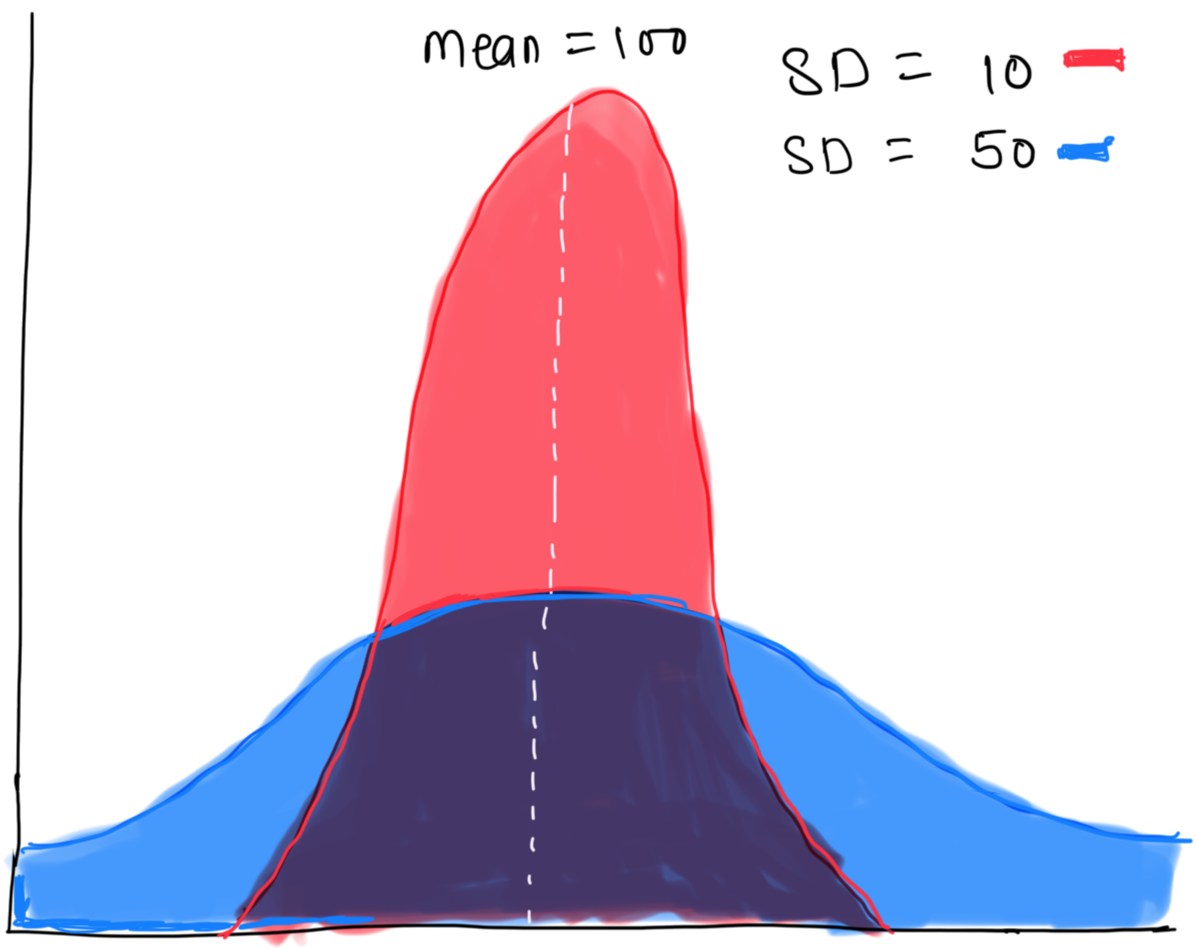
$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

Normal distribution (Bell Curve):

For normal data, ~68% falls within 1 SD

~95% —— " —— 2 SD

~99.7% —— " —— 3 SD of mean.



** Variance measure dispersion mathematically but is expressed in squared units, which makes interpretation difficult

** SD is square root of variance & is expressed in the same units as the data, making it easier to interpret

Where variance is used

Loss function
Bias-variance tradeoff
Model Theory

Where SD is used

Feature scaling explanations
EDA
Talking to non-technical stakeholders.

* Percentile and Quartiles

Percentile is a value below which a certain percentage of observations lie

eg It means that the person has got better marks than 95% of entire students

1 2 3 4 5 6 (index)
2, 2, 3, 4, 5, 5, 5, 6, 7, 8, 8, 8, 8, 8, 9, 9, $\boxed{10,}$ 11, 11, 12

What is percentile ranking of 10

$$= \frac{\#\text{of values below } 10}{\text{Total values}} \times 100$$

$$= \frac{16}{20} \times 100 = 80 \text{ percentile}$$

What value exists at percentile rank of 25

$$\text{Value} = \frac{\text{Percentile}}{100} \times (n+1)$$

$= \frac{25}{100} \times 21 = 5.25 \Rightarrow$ index

$\therefore$ value $= 5$ $\because$ $\frac{5+5}{2} = 5$ — from data

* Five number summary & Box plot

1) minimum
2) First Quartile (25%) Q_1
3) Median
4) Third Quartile (75%) Q_3
5) Maximum

Removing the outlier

eg $\{1, 2, 2, 2, 3, 3, 4, 5, 5, 5, 6, 6, 6, 6, 7, 8, 8, 9, 27\}$

Define $\rightarrow$ [Lower Fence $\longleftrightarrow$ Higher F]

$$\text{Lower fence} = Q_1 - 1.5\,(IQR)$$

$$\text{Higher Fence} = Q_3 + 1.5\,(IQR)$$

$$IQR = Q_3 - Q_1$$

$$Q_1 = \frac{25}{} \times (n+1), \quad Q_3 = \frac{75}{} \times (n+1)$$

100 100

= 3 = 7

IQR = 4 ∴ LF = −3 HF = 13

∴ Range [−3 ⟷ 13]

∴ 13 < 27 ∴ Outlier

1) min = 1
2) Q_1 = 3
3) median = 5
4) Q_3 = 7
5) Max = 9 (Don't take outlier)

Box Plot